Smart India Hackathon · Problem statement SIH26263 · AICTE

APAAR-Based Student Data Management and Verification System

Architecture and Implementation Blueprint



A national register of student enrolments for AICTE-approved institutions: student-level records keyed to APAAR ID, validated automatically, monitored against approved intake, and verifiable with consent.

Version 1.0 · 27 October 2026 **Status** Design baseline for the hackathon prototype

All student, institution and APAAR data referenced in this document is synthetic. APAAR registry verification in the prototype is simulated; no official API contract was available.

Contents

1 Executive summary 3
2 Problem statement and existing gap 4
3 Stakeholder analysis 5
4 Functional and non-functional requirements 5
5 Requirement traceability matrix 7
6 Research findings and benchmarking 8
7 Proposed solution and differentiators 10
8 Innovation and novelty 11
9 How the design solves the problem 12
10 Architecture decision: options compared 13
11 System context 15
12 Container and component architecture 16
13 Deployment architecture 18
14 Main data flow: from upload to dashboard 19
15 Student-record lifecycle 19
16 Bulk import and validation sequence 21
17 Cross-institution conflict resolution 23
18 Database design: entity relationships 24
19 Data dictionary, constraints and counting rules 24
20 API design and integration contracts 26
21 APAAR verification integration boundary 27
22 Credential issuance and verification 29
23 Security threat model 31
24 Privacy and authorization model 33
25 Dashboard and UX workflows 34
26 Validation and anomaly detection strategy 35
27 Scalability and performance 36
28 Observability, backup and recovery 36
29 Deployment and infrastructure plan 37
30 Testing and acceptance criteria 37
31 Prototype implementation scope (40–50%) 38
32 Limitations and external dependencies 39
33 Phased roadmap from prototype to production 39
34 Risk register 40
35 Assumptions and open questions 40
36 Abbreviations 41
37 References 41

Status labels used in this document: **Prototype** working in the prototype · **Simulated** working against a simulated dependency · **Planned** designed, not yet built. Bracketed numbers such as [3] refer to the references in the final section.

1 Executive summary

AICTE today receives detailed student-level data only from PGDM institutions; most other approved institutions report totals such as overall enrolment during the approval process. Totals cannot show who is enrolled, whether the same student is counted twice, or whether a programme admitted more students than it was approved for. SIH26263 asks for a platform that uses the APAAR ID as each student's academic identifier, collects student-level records from every institution after admissions, validates them automatically, and lets AICTE and authorised verifiers rely on them.

This blueprint proposes a **modular monolith** backed by PostgreSQL: one deployable application with strict internal module boundaries, a shared validation engine used by every intake channel (spreadsheet upload, institution-system API and manual entry), and a small set of background workers. It is the simplest design that is consistent, auditable and secure enough for a regulator, and it has a clear path to national scale without the operational cost of microservices on day one.

> **What sets the design apart.**
>
> 1. **Prevention, not only detection.** A live seat ledger and a seat-check API let an institution see, before confirming an admission, whether it would exceed its approved intake or a category cap. Excess admission becomes something the system helps avoid, not just something AICTE finds later.
> 2. **Seat accounting that follows the approval rules.** Approved intake, NRI seats and each supernumerary category (FN/OCI, CIWG, TFW) are counted separately, each cap records the rule it came from, and unused supernumerary seats are never silently transferred.
> 3. **Indian-name-aware identity resolution.** Probabilistic record linkage tuned for transliteration variants, initials and token order finds the same person under different spellings or IDs, and shows why it thinks so.
> 4. **Guided correction.** Every error says how to fix it; a mistyped APAAR ID gets a "did you mean" suggestion when exactly one near-variant matches the student's name and date of birth; one fix can be applied to all rows with the same problem.
> 5. **Explainable peer-group anomaly detection.** Each institution is compared with similar institutions using robust statistics; every point in its review-priority score has a named reason. A flag means "a person should look", never "fraud".
> 6. **Privacy by construction.** Cross-institution conflicts are visible to both institutions without revealing which other institution is involved; students share selective-disclosure credentials that reveal only the fields a verifier needs; employers need consent; students can raise correction requests on their own record.
> 7. **Tamper-evident audit and honest integration states.** A hash-chained audit log, pseudonymised APAAR IDs in logs, and every external integration labelled live, awaiting approval, simulated or not implemented.

The hackathon prototype implements roughly 40–50% of the target system as working vertical slices: authentication and role-based access, the registry and intake model, bulk import with the full rule catalogue and guided fixes, the seat-check API and live seat ledger, identity resolution, cross-institution conflict and excess-intake detection with a review workflow, institution and AICTE dashboards that update live, peer-group review priority, selective-disclosure credentials with tamper detection and revocation, consent-based

verification, student correction requests, CSV reports and the hash-chained audit trail. APAAR registry verification is simulated against a synthetic registry because no official API contract or credentials are available to participants; the adapter boundary is designed so a live integration replaces it without touching the rest of the system.

2 Problem statement and existing gap

Source requirements

The problem statement lists ten requirements, a target data flow (Institution → Student data upload → APAAR validation → Data verification → AICTE central repository → Analytics and monitoring → Authorised verification) and an expected outcome: authentic, verifiable student records linked to a national academic identity, real-time monitoring of admissions, detection of excess enrolment, and data-driven governance.

The gap today

Today	Consequence	What the platform changes
Aggregate enrolment reported during approval	Cannot see individual students; cannot detect duplicates or ghost enrolments	Student-level records keyed to APAAR ID, submitted after admissions
Different submission mechanisms by institution type	Inconsistent data, manual reconciliation	One template, one API, one rule set for all institutions
Intake checked against declared totals	Excess admissions and category misuse are hard to prove	Seat position computed from the records, per programme, year and category
No shared identifier across institutions	Same student can be enrolled at two institutions unnoticed	Cross-institution active-enrolment check on every commit
Verification by letters and emails	Slow for employers; forgeable	Signed records, QR verification, consent-based requests

Scope boundaries

The platform is a regulator's register of enrolments, not an admission system, fee system or learning-management system. It does not issue APAAR IDs (the national registry does) and does not replace the AICTE approval process; it consumes approved intake from it.

3 Stakeholder analysis

Each stakeholder's objective, pain point, data, permissions, typical error, risk and success measure. Permissions are enforced server-side (see the privacy and authorization model, page 33).

Stakeholder	Objective	Pain point today	Data they submit / use	Permissions	Common error or risk	Success measure
AICTE monitoring officers	Detect excess admissions and conflicts early; act on evidence	Unverifiable totals; manual follow-up by letter	Consume dashboards, flags, reports; request clarification	National read; resolve flags; set intake; revoke records	Acting on a false positive	Median time from flag to resolution; % programmes with complete submissions
AICTE policy analysts	Trends by state, discipline, nationality, category	No student-level data outside PGDM	Aggregates and exports	Read-only, masked identifiers	Re-identification from small cells	Reports used in policy notes without manual data collection
Institution nodal officers	Submit once, correctly, before the deadline; prove compliance	Repeated data requests in different formats	Submit records, declare completion, issue signed records	Own institution only	Submitting wrong seat categories; late changes	Upload-to-commit time; open flags per 100 students
Admission / data-entry staff	Get a spreadsheet accepted without guesswork	Rejected files with no row-level reasons	Upload and correct rows	Own institution; no credential issuance	Typos in APAAR ID, dates, codes	% rows accepted on first upload; corrections per upload
Students	Own an accurate record; control who verifies it	No visibility of what was reported	View record; consent or decline verification	Own record only	Misreported data; unwanted disclosure	Consent decisions recorded; correction requests resolved
Employers and external verifiers	Fast, trustworthy confirmation of enrolment	Fake certificates, slow letters	Present token or request with consent	Minimal fields after consent	Over-collection, enumeration of IDs	Verification time in minutes; zero disclosures without consent
Degree-awarding bodies	Print APAAR ID alongside student ID	No authoritative APAAR-to-roll mapping	Read verified mapping per student	Their affiliated institutions (roadmap)	Printing an unverified ID	% certificates carrying a registry-matched APAAR ID
Auditors	Show who did what, when, and that logs are intact	Mutable logs, no denied-access records	Audit trail, chain verification	Read-only audit	Undetected log tampering	Chain verification passes; all denials logged
Platform operators	Run reliably through admission-season peaks	Unplanned load spikes, integration failures	Metrics, logs, integration status	Ops roles, no data browsing	Insider misuse	Availability; recovery-point and recovery-time objectives met

4 Functional and non-functional requirements

Functional requirements

FR1–FR10 restate the official requirements. FR11–FR14 are inferred, with the reasoning shown so they can be challenged.

ID	Requirement	Source
FR1	Secure upload and update of student-level data keyed by APAAR ID, via web and API	Stated (R1)
FR2	Fields for approved intake, actual enrolment, supernumerary seats, NRI, OCI, FN status and other relevant student data	Stated (R2)
FR3	Automated detection of duplicates, invalid/missing APAAR IDs, excess admissions, inconsistencies, cross-institution duplicate enrolment	Stated (R3)
FR4	Real-time dashboards for AICTE and institutions	Stated (R4)
FR5	Secure verification for AICTE, institutions, employers, regulators	Stated (R5)
FR6	Verifiable student records linked to academic identity	Stated (R6)
FR7	Integration API for institutional and AICTE systems	Stated (R7)
FR8	Security, privacy, authentication, RBAC, audit trails, consent	Stated (R8)
FR9	Reports for policy, accreditation, monitoring, regulation	Stated (R9)
FR10	APAAR ID printed/displayed with student ID on certificates	Stated (R10)
FR11	Correction workflow with history; nothing silently overwritten	Inferred from FR3 and FR8
FR12	Students without an APAAR ID yet (e.g. foreign nationals) can be recorded with a documented reason	Inferred: APAAR registration relies on Aadhaar KYC [1]
FR13	Enrolment status changes (withdrawal, transfer, cancellation) that update seat counts	Inferred from FR3 (excess intake) and FR4
FR14	Submission-completion declaration per programme-year	Inferred from 'after the completion of admissions'

Non-functional requirements

Quality	Target or constraint
Security	OWASP ASVS L2 controls; TLS everywhere; secrets in a vault; signing keys in KMS/HSM
Privacy	Data minimisation; masked identifiers by default; consent for third-party verification; DPDP Act 2023 / Rules 2025 alignment to be confirmed by counsel [8]
Availability	99.5% target outside admission season, 99.9% during the submission window; graceful degradation when the APAAR registry is down
Performance	Validate a 5,000-row file in under 30 s; dashboard queries under 2 s at p95; verification under 500 ms at p95
Scalability	Sizing assumption: ~10,000 institutions, ~4 million new admissions per year, ~15 million active enrolments (to be validated with AICTE)
Auditability	Every write and every denied access recorded in a tamper-evident log, retained at least one year [8]
Interoperability	Versioned REST API, CSV/XLSX templates, W3C VC-shaped credentials [7], adapter boundary for national registries
Usability and accessibility	WCAG 2.1 AA; works on low-bandwidth connections; English first, Indian languages on the roadmap
Maintainability	One codebase, module boundaries enforced by import rules, rule catalogue as data, 80%+ test coverage on rules and permissions

5 Requirement traceability matrix

Every official requirement maps to a feature, module, entity, API, control, acceptance test and prototype status. Status values: Prototype working in the prototype, Simulated working against a simulated dependency, Planned designed, not built. Nothing is marked blocked except live APAAR access, which depends on an external approval.

Requirement	Feature	Module	Entities	API / workflow	Security control	Acceptance test	Prototype
R1 Secure upload/ update by APAAR ID	Bulk import with preview; manual entry; integration API	Ingestion, Records	ImportJob, ImportRow, Student, Enrolment	POST /imports, /imports/ {id}/commit, /enrolments, / integration/enrolments	AuthN, institution scope, file limits, idempotency	Upload demo file; commit; record appears only for owning institution	Prototype
R2 Approved intake	Intake per programme-year	Registry	ApprovedIntake	POST /intakes	AICTE_ADMIN only	Intake set; caps re-derived; capacity checks rerun	Prototype
R2 Actual enrolment	Seat position computed from records	Analytics	Enrolment (status, seat)	GET /programmes, / dashboard/*	Scope filters	Counts equal records with ACTIVE/PROVISIONAL	Prototype
R2 Supernumerary seats	Per-category caps with rule source	Registry, Validation	SupernumeraryQuota	GET /programmes	Configurable by AICTE	Boundary tests at cap, cap+1; non-transferable	Prototype
R2 NRI / OCI / FN status	Nationality status and seat category kept separate	Records	Student.nationality_status, Enrolment.seat_category	Template columns	Consistency rule R013	FN on Indian citizenship rejected; OCI on FN/OCI seat accepted	Prototype
R2 Other student information	Gender, DOB, citizenship, social category (sensitive), email	Records	Student	Template	Social category never shown to verifiers	Verifier response omits these fields	Prototype
R3 Duplicate records	In-file and in-repository checks; Indian-name-aware identity resolution with explained match weights	Validation	ValidationRule, ValidationIssue	Import preview	Non-destructive; review flags	F001, F002, F003, D002, D005 tests	Prototype
R3 Invalid/missing APAAR	Format check, registry check via adapter, pending reason	Validation, Adapters	Student.apaar_status	Import preview	Simulated adapter labelled	R003–R008 tests	Simulated
R3 Admissions exceeding intake	Category-aware ledger; flag after each commit/status change; seat-check API before admission	Validation, Review	ValidationIssue (C001, C002)	GET /issues	Human resolution only	Intake, intake+1, supernumerary excluded	Prototype

Requirement	Feature	Module	Entities	API / workflow	Security control	Acceptance test	Prototype
R3 Inconsistencies	Dates, programme ownership, category vs nationality, attribute conflicts	Validation	ValidationIssue	Import preview	Master record never overwritten	R010–R014, D004 tests	Prototype
R3 Cross-institution duplicate enrolment	Active-enrolment conflict on commit	Conflict detection	ValidationIssue (D003)	GET /issues/{id}	Other institution's identity withheld	Demo B; redaction test	Prototype
R4 Dashboards	AICTE national register, peer-group review priority, institution dashboard; live updates over server-sent events	Analytics	All	GET /dashboard/*	Role-scoped	Totals equal database counts	Prototype
R5 Secure verification	Selective-disclosure token verification; QR; consent-based requests	Verification	Credential, VerificationRequest	POST /verification/*	Rate limits, 2-identifier match, consent	Tamper, revoked, no-consent tests	Prototype
R6 Verifiable records	Ed25519-signed, SD-JWT-style selective-disclosure records, QR, revocation	Credentials	Credential	POST /enrolments/{id}/credentials, /.well-known/issuer-keys	KMS in production	Signature and revocation tests	Prototype
R7 Integration API	API key, scopes, idempotency; adapters for registry, intake feed, DigiLocker	Integration	IntegrationClient, IdempotencyRecord	POST /integration/enrolments	Scoped keys, replay-safe	Replay returns same result; body change = 409	Prototype (ERP push); others planned
R8 Security, RBAC, audit, consent	Role matrix, institution isolation, hash-chained audit, consent, student correction requests	IAM, Audit	User, AuditEvent	All	See threat model	Isolation, denial-audit, chain-tamper tests	Prototype
R9 Reports	Intake utilisation, nationality, flags, review priority, audit export	Analytics	All	GET /reports/*.csv	Exports audited; CSV-injection safe	Report totals reconcile	Prototype
R10 APAAR ID on certificates	Certificate view with APAAR ID + student ID + QR; data feed for degree bodies	Credentials	Credential	GET /credentials/{id}/certificate	Only registry-matched IDs	Certificate shows both IDs	Prototype (view); feed planned

6 Research findings and benchmarking

Sources were consulted on 27 October 2026. Primary sources are preferred; where only a secondary source was available it is marked, and the design makes the affected rule configurable rather than hard-coded.

Topic	Evidence	Design consequence	Basis
APAAR ID	Described as a 12-digit ID created through the Academic Bank of Credits with DigiLocker-based Aadhaar KYC; a PEN is required and the UDISE+ name must match Aadhaar [1]	Format rule is 12 digits with no invented checksum; registry check is a separate step; students without Aadhaar (e.g. foreign nationals) may have no APAAR ID, so a documented pending reason is supported	Secondary [1]
Supernumerary seats for international students	AICTE handbook clause 2.12.1(h): 15% supernumerary over approved intake for FN, OCI and CIWG, one-third reserved for CIWG [3]. The 2026-27 addendum reportedly raises the cap to 25% for PG engineering and for management and allied programmes, exclusively for international students [2]	Caps stored per intake with their rule source; percentages configurable by discipline and level; CIWG split modelled; conflicting editions flagged as an open question	Primary excerpt [3]; secondary [2]
NRI seats	Vacant approved-intake seats after the last state counselling round may be filled by NRI/FN/OCI/CIWG candidates with AICTE approval [3]	NRI counted within approved intake, separate from supernumerary	Primary excerpt [3]
Tuition Fee Waiver	Up to 5% of sanctioned intake, supernumerary; unfilled seats are not allotted to other categories [5]	TFW cap 5%; non-transferability enforced for all supernumerary categories	Secondary [5]
National Academic Depository	Only institutions upload awards, via CSV or API; verifiers register through API Setu; verification needs student consent except for statutory or investigating bodies; free of cost [6]	Same pattern: institution-only writes, registered verifiers, consent by default, statutory exception for AICTE	Primary [6]
Verifiable credentials	W3C VC Data Model 2.0 became a Recommendation on 15 May 2025; issuer/holder/verifier roles; secured with Data Integrity or JOSE/COSE [7]	Credential payload follows the VC 2.0 shape, signed as a compact JWS with Ed25519; conformance testing on the roadmap	Primary [7]
Data protection	DPDP Rules 2025 notified 14 Nov 2025; most obligations (notice, security safeguards, breach notice, retention) apply 18 months after notification; logs kept at least one year; verifiable parental consent for children [8]	Consent and notice built in; one-year minimum audit retention; legal basis for AICTE processing to be confirmed (open question)	Secondary legal summary [8]
SIH evaluation	The 2020 Ministry press release says domain experts from ministries, institutes and industry evaluated teams but publishes no scoring rubric [9]	No claims about judging criteria; the demo is built to show working end-to-end flows	Primary [9]

Comparable platforms

Platform	Verified capabilities	Pattern adopted	Gap relative to SIH26263
NAD / DigiLocker [6]	Institutions publish awards; consent-based verification; verifier onboarding	Institution-only writes; consent; registered verifiers	Holds awards, not in-progress enrolments or intake capacity
ABC / APAAR [1]	Lifelong academic ID; credit records	APAAR as the join key across institutions	Does not compute seat use against AICTE approval
W3C VC ecosystem [7]	Portable, issuer-signed claims	VC-shaped payload; public key publication	Needs a status/revocation mechanism and a regulator registry, which this platform provides

We do not claim knowledge of what other SIH teams build. Comparisons above are with public platforms only.

7 Proposed solution and differentiators

The platform is a **register**: institutions write enrolment records for their own students; the platform validates every record against the same rules regardless of channel; AICTE reads the national picture and reviews flags; students and verifiers interact through signed records and consent.

Differentiator	Why it matters for SIH26263	How it is measured
Rule-faithful seat accounting	Excess enrolment is the outcome AICTE asked for. Counting supernumerary seats against intake, or vice versa, produces false alarms or hides real excess	Boundary tests: 0, intake, intake+1, cap, cap+1; dashboard totals reconcile with records
Explainable rule catalogue	Regulators must justify actions; institutions must know how to fix rows	Every finding has rule ID, severity, message, evidence and remediation
Preview, correct, partial commit, retry	Thousands of institutions with uneven IT capacity must be able to finish submission	Share of rows accepted after correction; zero duplicates on retry
Privacy-preserving conflict detection	Cross-institution checks inherently reveal data about other institutions' students	Institution views never contain the other institution's name or programme
Consent-based, minimal verification	Employers need answers; students need control; enumeration of APAAR IDs must be impossible	Same response for match and non-match; disclosed field list is fixed
Tamper-evident audit	Disputes about who changed what must be settled from the log	Chain verification detects any edited or deleted event
Honest integration states	A demo that silently fakes national integrations damages credibility	Integration status endpoint and UI labels

8 Innovation and novelty

Innovation here means mechanisms that change outcomes for AICTE, institutions or students, not novelty for its own sake. The techniques (record linkage, robust statistics, digital signatures, hash chains) are individually established; the contribution is combining them around India's seat-approval model so that capacity, identity and disclosure are handled together. In our research we found no public platform that does this for AICTE-approved institutions: NAD holds awards, ABC/APAAR holds identities and credits, and neither computes seat use against approval [1][6].

Innovation	Common approach	What this design does	Why it matters	Prototype
1. Seat-check before admission and a live seat ledger	Excess intake is usually found after the fact, from annual returns	Institution admission software calls `POST /seat-check` (programme, year, seat category) before confirming a candidate and gets the position after admission: seats left within intake and in that category, or a clear 'this would exceed' answer. The same ledger drives the upload projection and pushes changes to dashboards within seconds over server-sent events.	Moves the regulator from detecting excess to helping prevent it; 'real-time monitoring' becomes literal	Prototype
2. Rule-faithful capacity ledger	Totals compared with a single intake number	Separate counters for within-intake (regular + NRI) and every supernumerary category, caps derived from a recorded rule source per programme-year, non-transferability enforced, CIWG one-third split modelled.	Avoids false alarms from supernumerary students and catches category misuse that totals hide	Prototype
3. Indian-name-aware identity resolution	Exact match on ID, or naive string similarity	Fellegi–Sunter-style record linkage over normalised name tokens (order-insensitive, initials expanded against full tokens, transliteration folding such as aa/a, ee/i, th/t, v/w, ph/f, double-consonant collapse), date of birth (exact, day/month swap), gender and institution history. Each candidate pair shows the agreement weights that produced its score.	Finds the same student enrolled twice under different spellings or IDs, which exact-ID checks miss, and explains every match	Prototype
4. Guided correction and bulk fixes	Upload rejected with a generic error	Every finding carries a remediation. A 12-digit APAAR ID that is not found is checked for single-digit substitutions and adjacent transpositions; if exactly one variant matches the student's name and date of birth the UI offers it as a suggestion (never applied automatically). 'Apply to all rows with this problem' fixes repeated errors such as a wrong programme code in one action.	Small institutions can finish submission without a helpdesk call; measured by rows accepted after correction	Simulated
5. Peer-group, explainable anomaly detection	Fixed thresholds or opaque ML scores	Robust z-scores (median and MAD) against peers of the same type, discipline and state for utilisation, supernumerary share, NRI share, late-change rate, share of admissions dated after the admission window, and date-of-birth clustering. Combined with rule signals into a review-priority score where every point names its reason.	Prioritises AICTE's limited review capacity without accusing anyone; no labelled data is needed	Prototype
6. Privacy-preserving conflict detection	Either no cross-institution check, or full disclosure to every institution	Conflicts are detected on keyed pseudonyms of APAAR IDs; each institution sees that a conflict exists and its own side only; AICTE sees both sides.	Enables the cross-institution check the problem asks for without creating a new leak	Prototype

Innovation	Common approach	What this design does	Why it matters	Prototype
7. Selective-disclosure credentials	PDF certificates or all-or-nothing signed records	Records issued in the SD-JWT style: each claim is a salted digest inside the signed payload; the student discloses only the claims a verifier needs (for example institution, programme and status, but not the full APAAR ID). Verified offline against published keys, with online revocation status.	Data minimisation at the point of verification, aligned with the consent model of NAD and DPDP	Prototype
8. Student-centred correction loop	Students cannot see or challenge what institutions report	Students see their record, approve or decline verification requests, and raise a correction request that lands in the institution's queue with a due date and appears in AICTE's review signals if ignored.	Turns the people most likely to notice an error into a data-quality channel; supports the right to correction	Prototype
9. Offline-friendly submission	Online forms that fail on poor connections	Excel template with drop-down lists and in-sheet checks; the same rule definitions (exported as JSON) run in the browser to pre-check a file before upload; only the result is sent.	Works for institutions with limited bandwidth; fewer round-trips	Planned
10. Tamper-evident audit	Mutable application logs	Hash-chained audit events with pseudonymised identifiers, denied-access events included, periodic anchoring of the chain head to write-once storage.	Disputes about who changed a record are settled from evidence	Prototype

What is genuinely novel, stated plainly

- **Prevention at the point of admission** through a regulator-run seat ledger that understands seat categories.
- **Identity resolution tuned to Indian naming**, with explanations attached to every match, applied across institutions on pseudonymised identifiers.
- **Disclosure that matches the purpose**: institutions see their side of a conflict, verifiers see only consented, selectively disclosed claims, analysts see masked identifiers.

Everything else is careful engineering of known patterns, which is what a regulator should expect from a system it will depend on.

9 How the design solves the problem

The problem statement names five expected outcomes and an innovation expectation. The table shows the mechanism that delivers each outcome and how success would be measured in a pilot.

Expected outcome	Mechanism in this design	How it is measured
Authentic records linked to a national academic identity	APAAR ID as the join key; registry check through the adapter; identity resolution for spelling variants; master identity never overwritten by a second institution	Share of active students with a registry-matched APAAR ID; duplicate-identity candidates resolved
Student-level data from every institution after admissions	One template, Excel/CSV upload with guided fixes, ERP API with idempotency, completion declaration per programme-year	Programmes with a declared submission by the deadline; median upload-to-commit time

Expected outcome	Mechanism in this design	How it is measured
Real-time monitoring of admissions	Live seat ledger updated on every commit and status change; dashboards pushed over server-sent events; seat-check API used during admission	Seconds from commit to dashboard; seat checks per admission
Detect excess enrolment	Category-aware counting rules, C001/C002 flags with evidence, prevention through seat-check	Excess seats flagged; excess seats avoided after seat-check adoption
Detect duplicates and invalid/missing APAAR IDs	25-rule catalogue across row, file, repository and aggregate stages; cross-institution conflict detection; 'did you mean' repair	Rule hit rates; share fixed before commit
Secure verification for AICTE, institutions, employers, regulators	Signed selective-disclosure records, QR verification, consent-based requests, statutory access for AICTE	Verification time; disclosures without consent (target zero)
Data-driven governance and policy	National register by state, discipline, nationality and category; peer-group review priority; CSV reports for accreditation and monitoring	Reports used without manual data calls
APAAR ID on certificates	Certificate view and degree-body feed pairing institution student ID with a verified APAAR ID	Certificates carrying a registry-matched APAAR ID

The target flow, end to end

Institution → Student data upload: template, Excel/CSV or ERP API, pre-checked in the browser. → **APAAR validation:** format check, registry check through the adapter, guided repair. → **Data verification:** 25 rules, identity resolution, cross-institution conflict check, capacity ledger. → **AICTE central repository:** versioned records with revisions and a hash-chained audit trail. → **Analytics and monitoring:** live national register, peer-group review priority, reports. → **Authorised verification:** selective-disclosure signed records, QR, consent-based requests.

10 Architecture decision: options compared

Two viable options were evaluated. Scores are relative (more dots is better for the platform's needs).

Criterion	A. Modular monolith + PostgreSQL + workers	B. Microservices (separate ingestion, validation, records, credentials, analytics services) with event bus
Implementation speed	●●● One codebase, one deploy, shared engine	● Service contracts, messaging, distributed tracing before features
Data consistency	●●● Commit, revision, flag and audit in one ACID transaction	● Sagas or outbox per service; audit chain spans services
Operational cost	●●● Few moving parts; small ops team	● Many deployables, a broker, service mesh
Scalability	●● Horizontal stateless app tier; read replica; partitioning; extract hot modules later	●●● Independent scaling per service
Failure isolation	●● Bulkheads by worker pools and timeouts; adapter circuit breakers	●●● Process-level isolation
Security	●●● One authorization layer, fewer network paths	●● More surfaces; service-to-service auth needed

Criterion	A. Modular monolith + PostgreSQL + workers	B. Microservices (separate ingestion, validation, records, credentials, analytics services) with event bus
Maintainability	●● Requires discipline on module boundaries	●● Clear ownership, but duplicated cross-cutting code

Decision: option A, with planned extraction points

The workload is write-bursty during admission season and read-heavy otherwise; the hard requirement is consistency between a record, its flags and its audit event. A modular monolith delivers that in one transaction. Scale comes from stateless app replicas, a read replica for analytics, background workers for slow work, and table partitioning by admission year. Two modules are designed for extraction if load demands it: Verification (public, latency-sensitive traffic) and Analytics (heavy reads). Their interfaces are already narrow.

Rejected elsewhere: blockchain (a hash-chained log plus signed credentials gives tamper evidence without consensus overhead; there is no multi-party write problem to solve), Kubernetes at prototype stage (containers on autoscaled VMs are enough), and ML anomaly models (no labelled ground truth exists; rules and transparent statistics come first).

Risks of the choice: module boundaries can erode (mitigated by import-linting and module-owned tables), and a single deployable means a bad release affects all functions (mitigated by blue-green deploys and feature flags).

11 System context

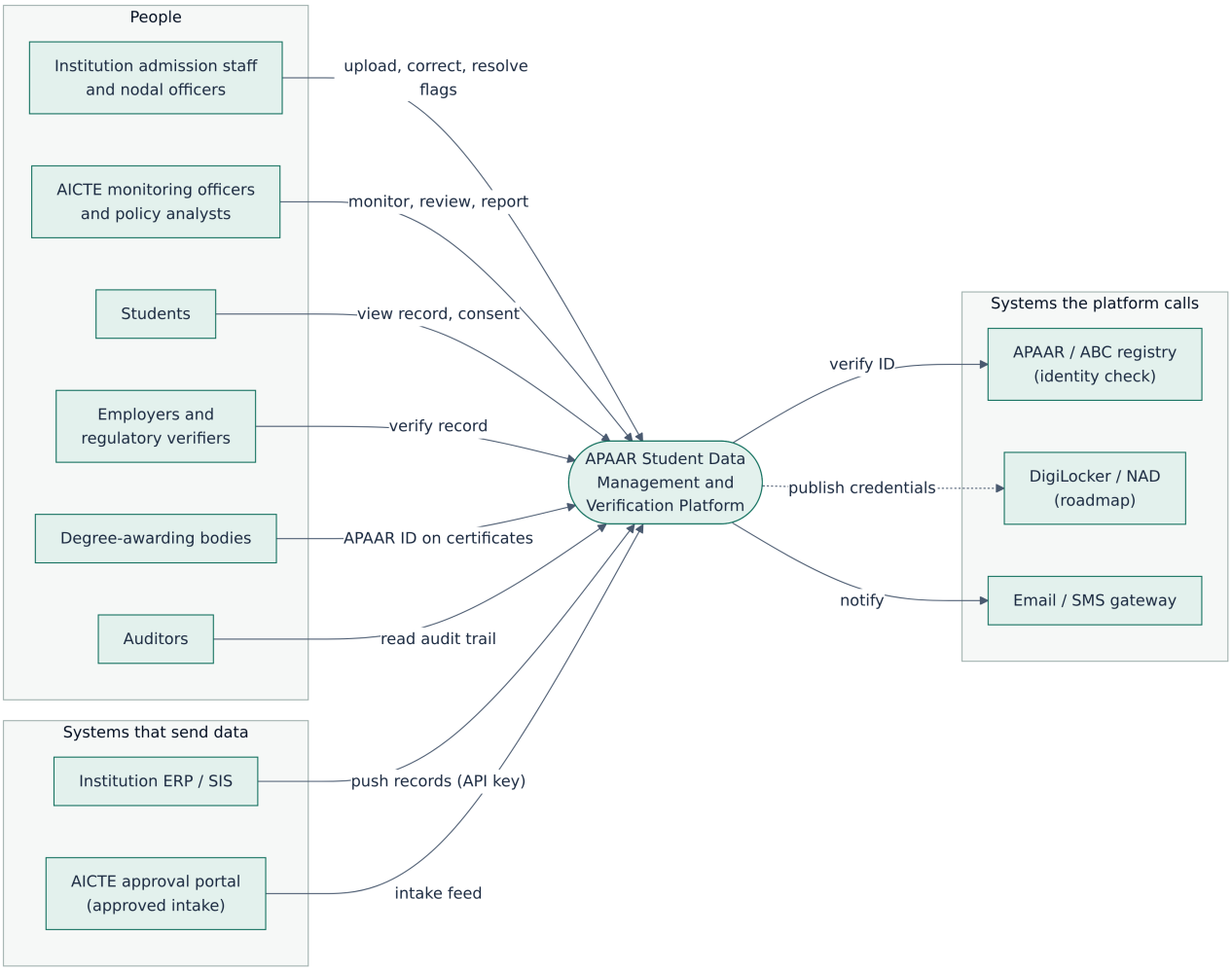


Figure 1. System context. Solid lines are in scope for the target system; the dotted DigiLocker link is a roadmap item. In the prototype the APAAR registry is simulated and the intake feed is entered by AICTE administrators.

12 Container and component architecture

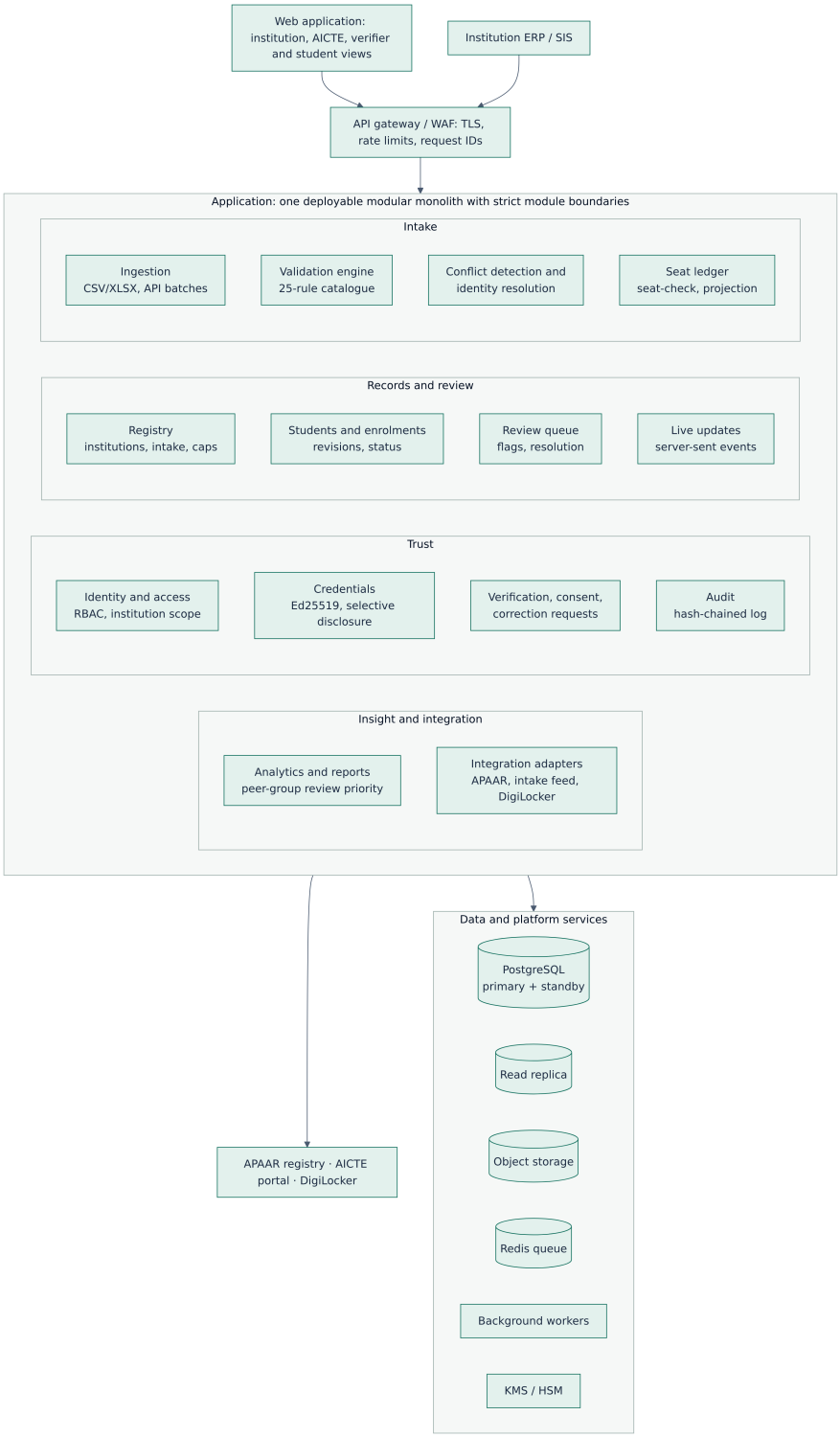


Figure 2. Containers and internal modules. Modules communicate through in-process interfaces; only the adapters cross the network to external systems.

Module responsibilities and boundaries:

Module	Owns	Exposes	Must not
Identity and access	Users, roles, sessions, API clients	Principal, role and scope checks	Contain business rules

Module	Owns	Exposes	Must not
Registry	Institutions, programmes, approved intake, quotas	Intake and caps for a programme-year	Count enrolments itself
Ingestion	Import jobs and rows, file parsing	Preview, correct, commit	Write enrolments except through the engine
Validation engine	Rule catalogue and evaluation	validate(rows), commit(values), aggregate checks	Depend on HTTP or UI
Conflict detection and identity resolution	Cross-institution checks; record linkage with match weights	Findings and candidate pairs with evidence	Reveal other institutions to institution users; merge identities automatically
Seat ledger	Live capacity position per programme-year and category	Seat-check, projection, change events	Block a commit (it informs; AICTE decides)
Records	Students, enrolments, revisions, status changes	Search, detail, correction	Overwrite master identity silently
Review queue	Issues and their event history	Actions by role	Auto-resolve except cleared capacity conditions
Credentials	Signing, revocation, certificate view	Issue, revoke, verify	Hold private keys outside KMS in production
Verification and consent	Verification requests, consent decisions, student correction requests	Token verify, consent flow, correction queue	Disclose more than the fixed field list
Live updates	Server-sent event streams per role scope	Seat and flag change events	Push data outside the subscriber's scope
Analytics and reports	Read models and exports	Dashboards, CSV	Write to transactional tables
Integration adapters	APAAR, intake feed, DigiLocker clients	Typed results with mode label	Fall back silently to simulated data
Audit	Hash-chained event log	record(), verify_chain()	Store raw APAAR IDs, names or DOB

13 Deployment architecture

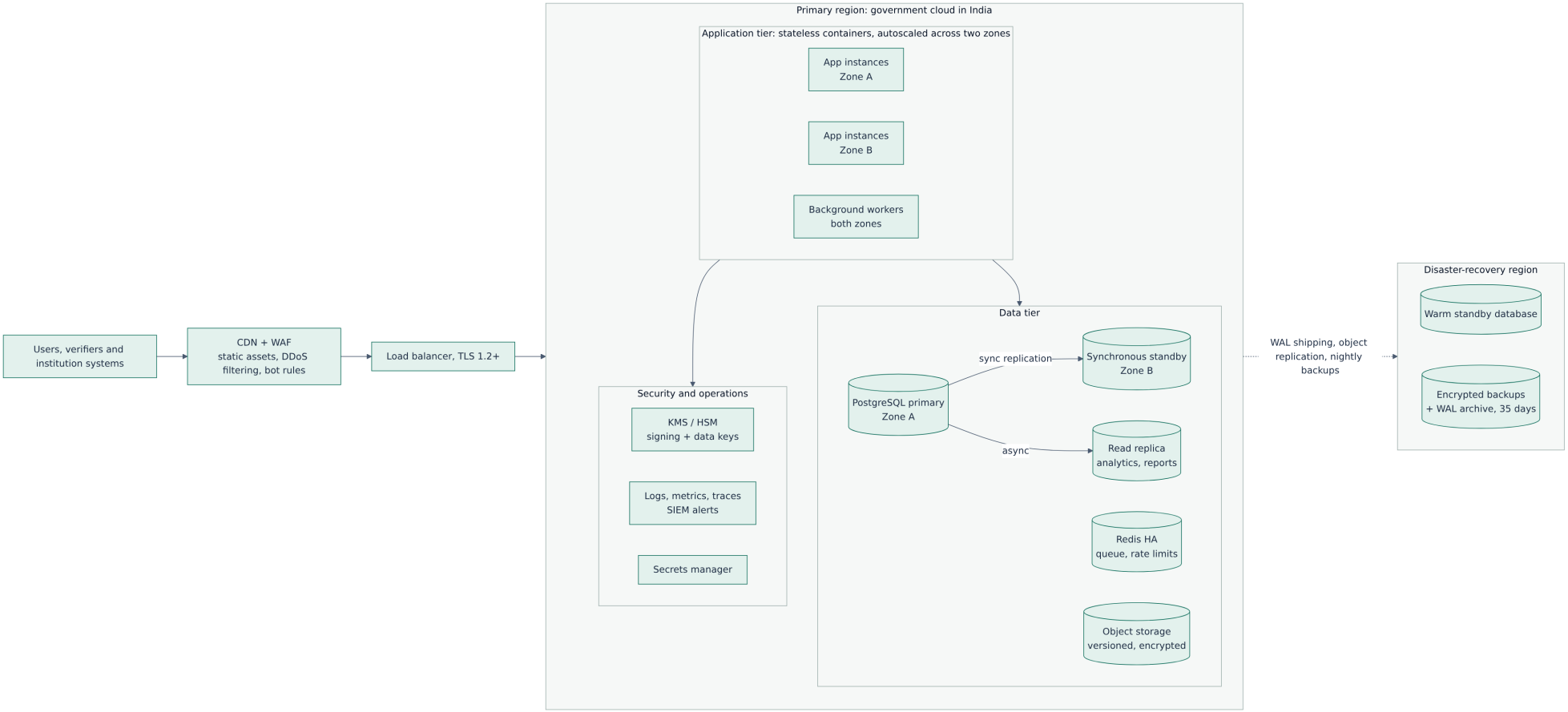


Figure 3. Production deployment on a government cloud in India, two availability zones plus a disaster-recovery region.

The app tier is stateless, so instances scale horizontally behind the load balancer. PostgreSQL runs with a synchronous standby in a second zone (no data loss on zone failure) and an asynchronous read replica for dashboards and exports. Uploaded files and generated reports live in versioned, encrypted object storage. Signing and data-encryption keys stay in KMS/HSM. The DR region receives continuous WAL shipping and nightly encrypted backups.

14 Main data flow: from upload to dashboard

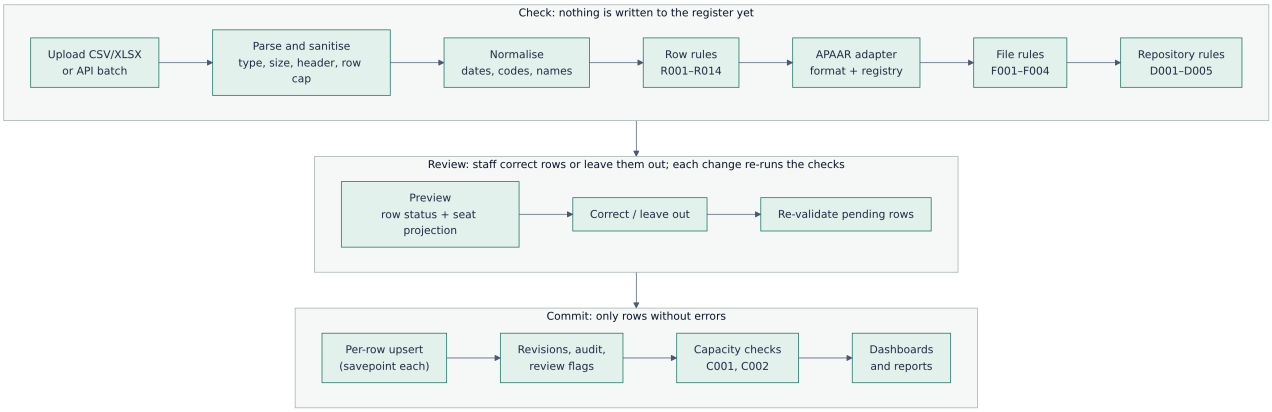


Figure 4. Ingestion data flow. The same pipeline serves spreadsheet uploads, the institution-system API and manual entry.

Three properties matter most. First, **nothing reaches the register until commit**; a preview is a sandbox the institution can correct freely. Second, **the commit re-validates**, because another institution may have committed a conflicting record between preview and commit. Third, **each row commits in its own savepoint**, so one bad row cannot roll back a batch of good ones, and committed rows are never committed twice.

Verification data flow

Signed records flow out of the register to students; verifiers flow back in either by presenting a token (holder-initiated, so presentation is consent) or by requesting with two identifiers and waiting for consent. Both paths are rate-limited and audited (Figure 9).

15 Student-record lifecycle

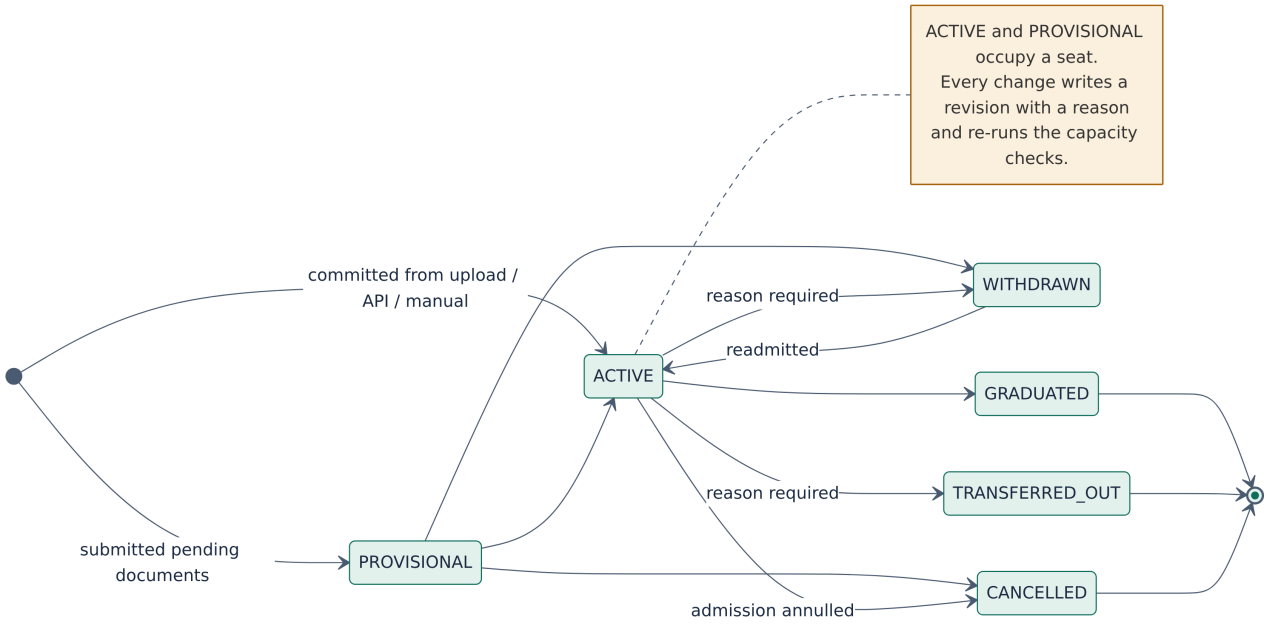


Figure 5. Enrolment status transitions. Disallowed transitions are rejected by the API.

Record	Created by	Changed by	Never
Student (identity)	First commit of an APAAR ID	Owning institution correction with reason; APAAR added when issued	Overwritten by a different institution's upload (raises D004 instead)
Enrolment	Commit from upload, API or manual entry	Correction with reason; status transition with reason	Hard-deleted; edited without version check
Validation issue	Engine on commit / capacity check	Review actions by role	Deleted; auto-closed except when a capacity condition clears (logged as system event, reopenable)
Credential	Nodal officer issue	Revocation (institution or AICTE); superseded on re-issue	Re-signed in place

16 Bulk import and validation sequence

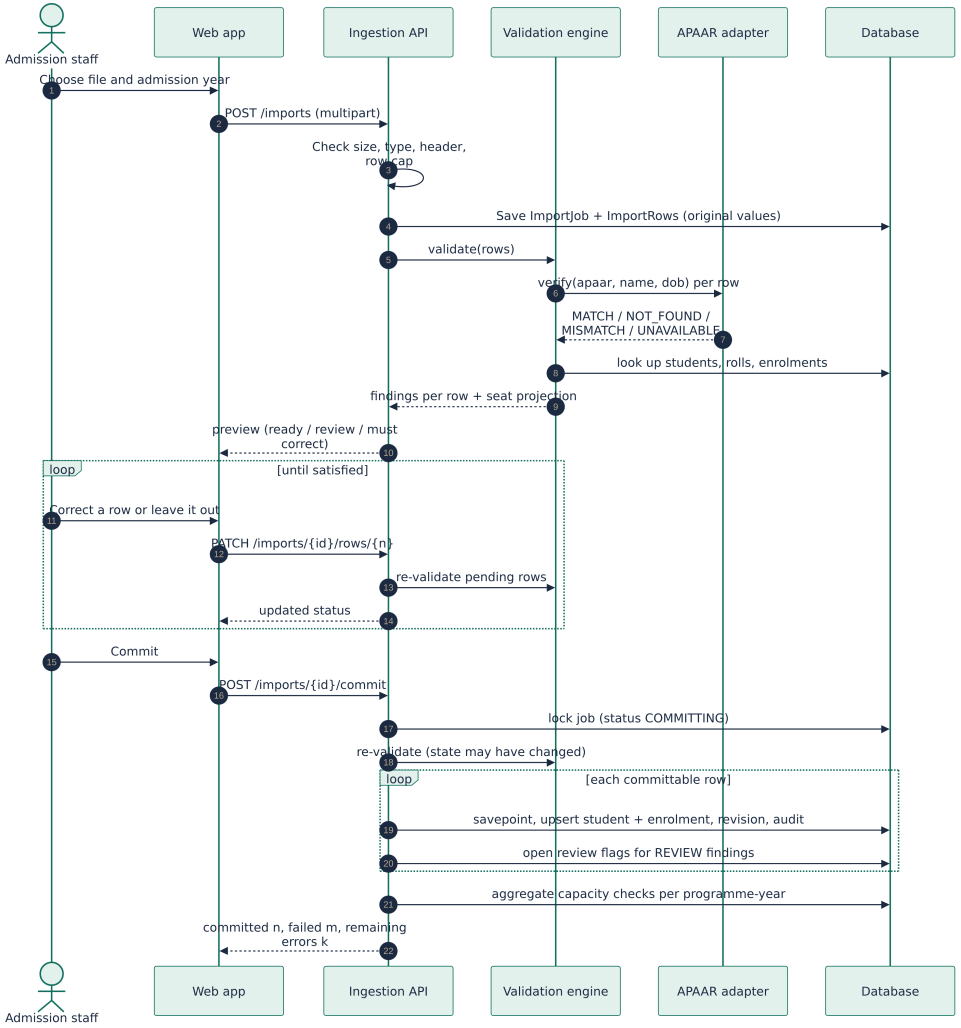


Figure 6. Bulk import sequence, including the correction loop and the locked, re-validated commit.

Recovery behaviour. If the registry adapter is unreachable the row gets a warning (R008) and is stored UNVERIFIED for automatic re-verification. If a commit crashes midway, committed rows are marked and the job is PARTIALLY_COMMITTED; the next commit picks up only remaining rows. A second concurrent commit of the same job is refused (409). Uploading the identical file again is flagged (F004) and updates rather than duplicates.

17 Cross-institution conflict resolution

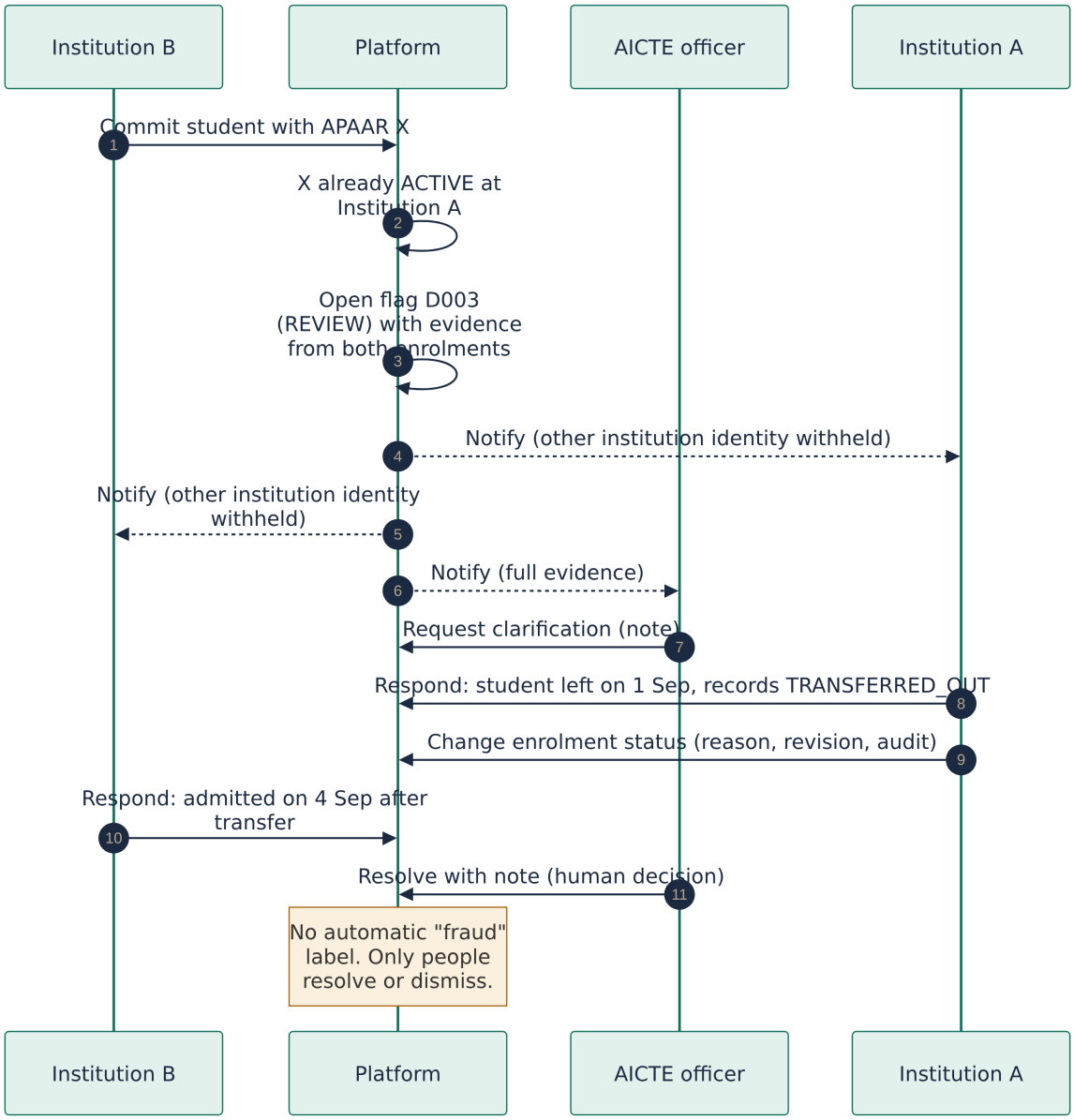


Figure 7. Resolving an APAAR ID that is active at two institutions.

A cross-institution conflict has several innocent explanations: a transfer not yet recorded, a permitted dual registration, or a typo in the APAAR ID. The platform therefore opens one review flag linked to both institutions, shows each institution only its own side, and gives AICTE the full evidence. Resolution is always a human decision with a note. The same principle applies to excess-intake flags: the records are accepted (blocking them would reward under-reporting), and the flag clears automatically only if the measured condition disappears, for example after an institution records withdrawals; that system event is logged and reopenable.

18 Database design: entity relationships

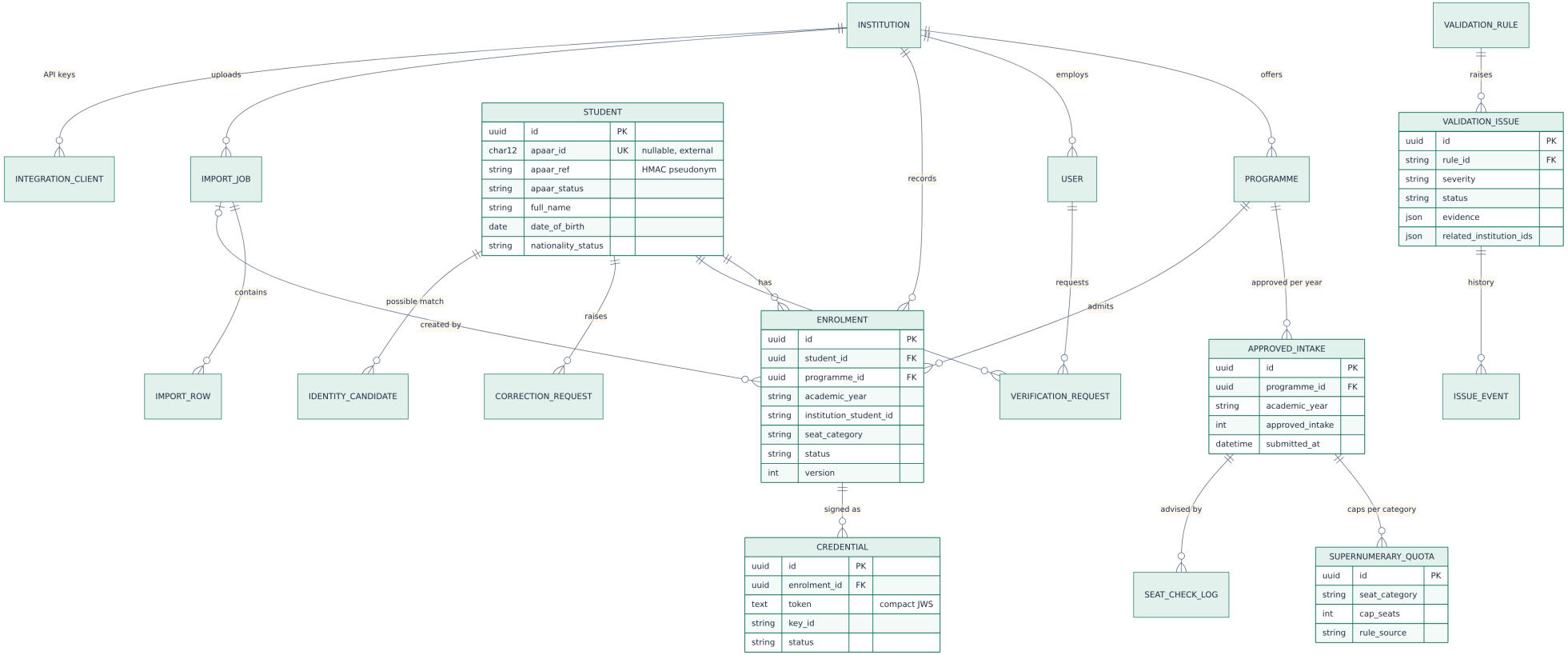


Figure 8. Core entities. Full column lists are in the data dictionary that follows.

19 Data dictionary, constraints and counting rules

Entity	Key columns and constraints	Notes
Institution	PK id (UUID); UNIQUE aicte_code	State and region drive national filters

Entity	Key columns and constraints	Notes
Programme	PK id; UNIQUE (institution_id, code)	Level DIPLOMA/UG/PG and discipline select the supernumerary percentage
ApprovedIntake	PK id; UNIQUE (programme_id, academic_year); approved_intake ≥ 0	submitted_at records the completion declaration
SupernumeraryQuota	UNIQUE (intake_id, seat_category); cap_seats ≥ 0; rule_source NOT NULL	Caps derived from rules, overridable by AICTE with an audit event
Student	PK id; UNIQUE apaar_id WHERE NOT NULL; INDEX (name_key, date_of_birth); apaar_ref HMAC pseudonym	APAAR ID is external, never a primary key, never in URLs or logs
Enrolment	PK id; UNIQUE (institution_id, institution_student_id); UNIQUE (student_id, programme_id, academic_year); INDEX (programme_id, academic_year, status); version for optimistic locking	One student may have enrolments in several years and, after transfer, institutions
ImportJob / ImportRow	UNIQUE (job_id, row_no); file_sha256 indexed	Original values kept beside corrected values
ValidationRule / ValidationIssue / IssueEvent	Issue INDEX (rule_id, dedupe_key); one unresolved issue per rule and key	Production adds an issue_institution join table instead of a JSON list
RecordRevision	INDEX entity_id; (entity_id, version) unique	Field-level old/new values with reason and actor
AuditEvent	PK seq; prev_hash, hash NOT NULL	Append-only; database role has INSERT and SELECT only
Credential	PK id; INDEX enrolment_id; status ACTIVE/REVOKED/ SUPERSEDED	Token stored for re-verification by reference
VerificationRequest	INDEX requester_id, student_id; expires_at	NO_MATCH stored but shown to the verifier as pending
CorrectionRequest	PK id; INDEX (institution_id, status); due_at	Raised by a student on their own record; resolution recorded with a note
IdentityCandidate	PK id; UNIQUE (student_a, student_b); score; weights JSON; decision	Output of identity resolution and reviewer decisions
SeatCheckLog	PK id; INDEX (programme_id, academic_year)	Advisory checks requested before admission
IntegrationClient / IdempotencyRecord	UNIQUE key_prefix; key_hash only; PK (client:key)	Keys shown once at creation

Counting rules (the definition of "excess")

1. Only enrolments with status ACTIVE or PROVISIONAL occupy a seat.
2. REGULAR and NRI seats count against the approved intake. NRI seats are filled against vacant approved-intake seats [3].
3. Each supernumerary category (SN_FN_OCI, SN_CIWG, SN_TFW, SN_OTHER) counts only against its own cap, never against the approved intake.
4. Unused supernumerary seats are not transferable: a category over its cap is flagged even if another has spare seats [5].

5. Caps are whole seats, rounded down from percentages (assumption; open question Q4). FN/OCI/CIWG pool: 15% for diploma and UG engineering, 25% for PG engineering and for management and allied programmes [2], split one-third CIWG and two-thirds FN/OCI [3]. TFW: 5% [5].
6. Excess over intake = max(0, within-intake enrolled − approved intake). Category excess = max(0, enrolled − cap).

Transfers, withdrawals and history

A transfer is two facts: TRANSFERRED_OUT at the old institution and a new enrolment at the receiving one. Withdrawals free the seat from the date recorded. Nothing is deleted; every change has a revision. Retention: enrolment records for the life of the academic identity; import files 3 years; audit events at least 1 year online [8] and 7 years archived (assumption to confirm). Tables are partitioned by admission year in production.

Concurrency and transactions

One transaction per committed row (savepoint) plus one for capacity checks; optimistic version checks on corrections and status changes; a job-level lock prevents parallel commits; the audit chain head is serialised with a database advisory lock in PostgreSQL.

20 API design and integration contracts

Group	Endpoints (v1)	Who
Auth	POST /auth/login; GET /auth/me	All users
Registry	GET /institutions, /programmes; POST /intakes; POST /intakes/{id}/submit	AICTE; institutions (own)
Imports	GET /imports/template.csv, /imports/template.xlsx (drop-downs); POST /imports; POST /imports/{id}/bulk-fix (field, from, to); GET /imports/{id}; PATCH /imports/{id}/rows/{n}; POST /imports/{id}/rows/{n}/exclude; POST /imports/{id}/commit; POST /imports/{id}/discard; GET /imports/{id}/report.csv	Institution staff
Records	GET /enrolments (search, filter, sort, page); GET /enrolments/{id}; PATCH /enrolments/{id}; POST /enrolments/{id}/status; POST /enrolments	Institutions (own); AICTE (all)
Review	GET /issues; GET /issues/{id}; POST /issues/{id}/actions; GET /issues/export.csv	AICTE; institutions (own); auditors
Seat ledger	POST /seat-check (programme, year, seat category, count); GET /events (server-sent events: seat and flag changes in the caller's scope)	Institutions (own), their ERPs; AICTE
Dashboards and reports	GET /dashboard/national, /dashboard/institution, /dashboard/review-priority; GET /reports/*.csv	AICTE; institutions (own)
Identity resolution	GET /identity/candidates (pairs with match weights); POST /identity/candidates/{id}/decision (same person / different people)	AICTE; institutions for own students
Students	GET /me/record; POST /me/correction-requests; GET /correction-requests (institution queue); POST /correction-requests/{id}/actions	Students; institutions
Credentials	POST /enrolments/{id}/credentials (returns SD-JWT with disclosures); POST /credentials/{id}/revoke; GET /credentials/{id}/certificate, /qr.svg; GET /.well-known/issuer-keys	Nodal officers; AICTE; public keys open
Verification	POST /verification/verify-token; GET /verification/credentials/{id}; POST/GET /verification/requests; POST /verification/requests/{id}/consent; GET /verification/requests/{id}/result	Public (rate-limited); verifiers; students
Integration	POST /integration/enrolments (X-API-Key, Idempotency-Key); GET /integration/status	Institution systems
Audit and rules	GET /audit, /audit/verify-chain, /audit/export.csv; GET/PATCH /rules	AICTE admin, auditors

Conventions

- **Versioning:** URL major version (/api/v1); additive changes only within a version; deprecation header and six-month overlap.
- **Errors:** `{"error": {"status", "message", "request_id", "fields"|"findings"}}`; no stack traces; 404 instead of 403 for objects outside the caller's institution so existence is not disclosed.
- **Pagination:** page and page_size (max 200) with total and pages; cursor pagination for exports at scale.
- **Idempotency:** integration writes require Idempotency-Key; same key and body returns the stored response; same key with a different body returns 409.
- **Documentation:** OpenAPI generated from the code and published at /api/docs.

Example: institution-system push

```
POST /api/v1/integration/enrolments
X-API-Key: sk_xxxx.********      Idempotency-Key: 2026-08-14-batch-007
{"academic_year": "2026-27", "records": [{"apaar_id": "9xxxxxxxxxxx", "institution_student_id":
"KIT26CSE042",
  "full_name": "...", "date_of_birth": "2008-04-15", "gender": "F", "nationality_status": "INDIAN",
  "programme_code": "BTECH-CSE", "seat_category": "REGULAR", "admission_date": "2026-08-01"}]}

200 {"accepted": 1, "rejected": 0, "results": [{"index": 0, "status": "VALID", "committed": true,
     "enrolment_id": "...", "findings": []}]}
```

Example: seat check before confirming an admission

```
POST /api/v1/seat-check   {"programme_code": "BTECH-CSE", "academic_year": "2026-27",
"seat_category": "NRI", "count": 1}
200 {"allowed_within_rules": false, "category": "NRI", "counts_against": "approved intake",
     "within_intake": {"approved": 60, "occupied": 60, "after": 61}, "message": "Admitting would
exceed the
     approved intake by 1. NRI seats are filled against vacant approved-intake seats."}
```

The seat check is advisory: it never blocks an admission, because the institution and AICTE remain the decision makers. Every check is logged so later excess can be compared with the advice given.

21 APAAR verification integration boundary

All identity checks go through one adapter interface: `verify(apaar_id, name, date_of_birth) → MATCH | NOT_FOUND | NAME_MISMATCH | DOB_MISMATCH | INACTIVE | UNAVAILABLE`, each result carrying a `mode` label. No official APAAR or ABC API contract was available, so none is invented here.

State	Meaning	Prototype
LIVE	Authorised integration with the national registry (mutual TLS, client credentials, contract from the registry owner)	Not available
AWAITING	Adapter written against a published contract, waiting for credentials or approval	Not available
SIMULATED	Checks against a synthetic registry table seeded with synthetic identities; every result and UI label says so	In use

State	Meaning	Prototype
NOT_IMPLEMENTED	Declared but not built	Intake feed, DigiLocker

- **A well-formed ID is not an identity.** Format (12 digits) and registry match are separate rules (R005, R006/R007).
- **Degraded mode.** If the registry is down, rows are accepted as UNVERIFIED with a warning, and a worker re-verifies with exponential back-off. A circuit breaker stops hammering a failing registry.
- **Data minimisation.** Only APAAR ID, name and date of birth are sent; responses are stored as status plus timestamp, not as copies of registry data.
- **No silent substitution.** Production configuration refuses to start with the simulated adapter.

22 Credential issuance and verification

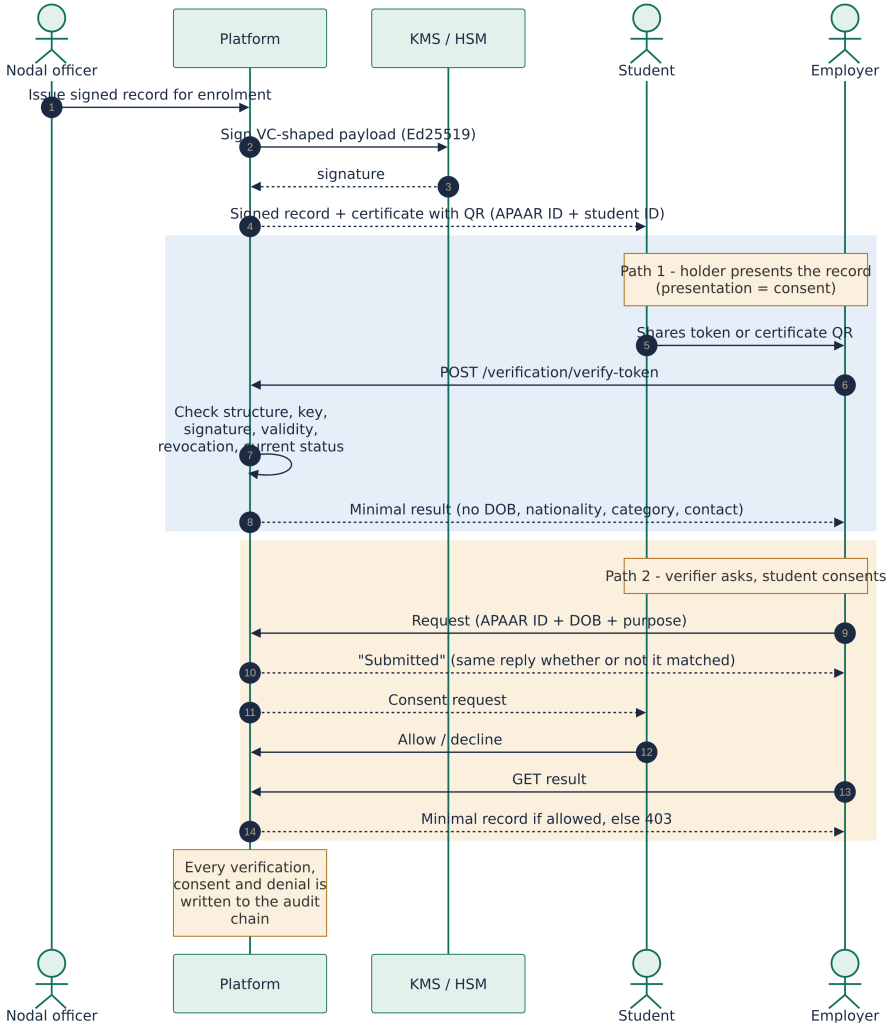


Figure 9. Issuing a signed record and the two verification paths.

- **Format:** W3C VC 2.0-shaped JSON [7] (context, type StudentEnrolmentRecord, issuer, validFrom/validUntil, credentialSubject, credentialStatus) signed as a compact JWS, alg EdDSA (Ed25519), kid in the header.

- **Selective disclosure:** following the IETF SD-JWT approach, each subject claim is replaced in the signed payload by the SHA-256 digest of a salted disclosure. The student shares the token plus only the disclosures a verifier needs (for example institution, programme and status); the verifier recomputes digests and checks the signature. Undisclosed claims stay hidden but cannot be altered. In the UI the student ticks which fields to share.
- **Subject fields:** APAAR ID, name, institution student ID, institution and AICTE code, programme, admission year, status at issuance. No date of birth, category or contact data.
- **Keys:** private key in KMS/HSM; public keys at /.well-known/issuer-keys (JWKS). Rotated keys stay published so older records still verify.
- **Checks on verification:** structure, known key, signature, validity period, revocation/supersession, and a currency note if the enrolment status changed after issuance.
- **Revocation:** online status lookup in the prototype; a W3C Bitstring Status List is the planned scalable, privacy-preserving replacement.
- **Certificates (R10):** the certificate view prints the institution student ID next to the APAAR ID with a QR code to the verification page; degree-awarding bodies receive the verified APAAR-to-roll mapping through the API (planned).

23 Security threat model

STRIDE categories: **S**poofing, **T**ampering, **R**epudiation, **I**nformation disclosure, **D**enial of service, **E**levation of privilege.

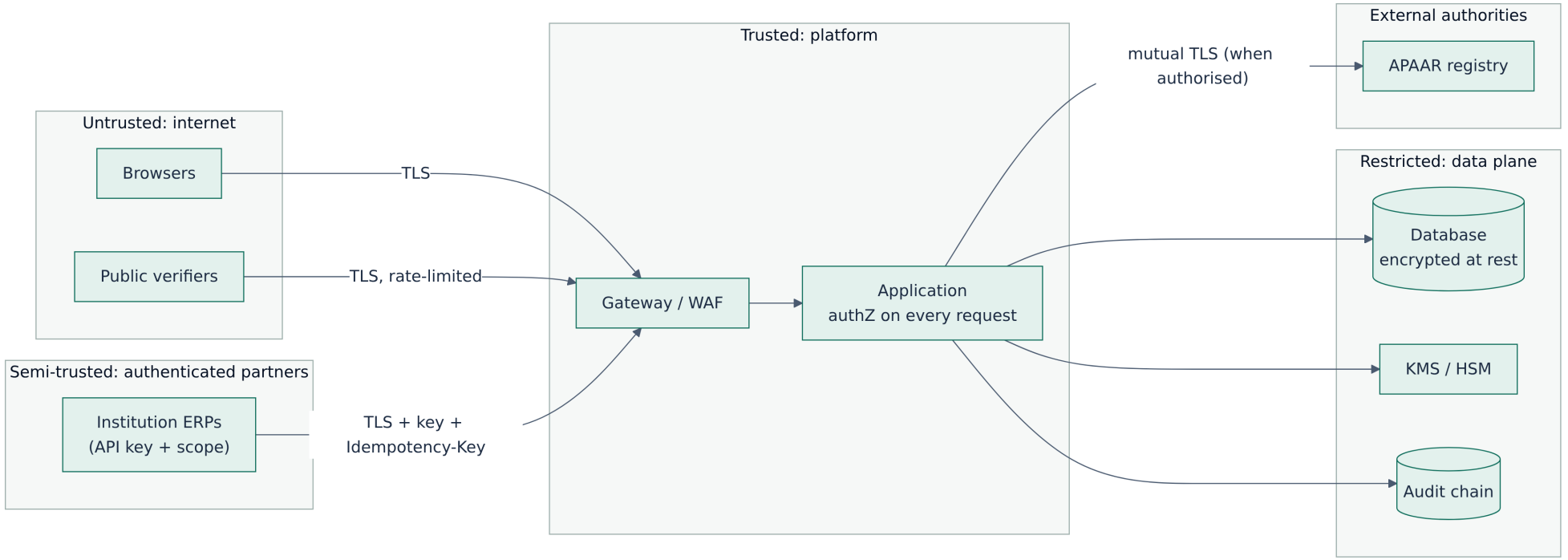


Figure 10. Trust boundaries. Authorization is enforced inside the application on every request, not at the gateway alone.

Threat	STRIDE	Control	Verification
Cross-institution data leakage	I	Institution scope applied in every query; 404 for foreign objects; redacted conflict evidence	Isolation tests on list, detail, write, export
Broken object-level authorization	E	Ownership check on every object route, not only on lists	Tests per route
Privilege escalation	E	Role from server-side user record, not token claims alone; admin actions restricted	Role matrix tests
Credential theft / session hijack	S	Short-lived tokens, PBKDF2 password hashing, lockout, MFA for AICTE and nodal officers (planned)	Login and lockout tests

Threat	STRIDE	Control	Verification
Forged or replayed imports	S/T	Authenticated upload; file hash recorded; idempotency keys on API	Replay test
Duplicate or concurrent requests	T	Job lock, savepoints, optimistic versions	Concurrency tests
API abuse and ID enumeration	I/D	Rate limits; two-identifier match; identical response for match/no match; no partial APAAR search	Rate-limit and enumeration tests
Tampered credentials	T	Ed25519 signatures; revocation check	Tamper test
Excessive data exposure	I	Masked APAAR by default; fixed verifier field list; social category never disclosed	Response-shape tests
Insider misuse	R/I	Every record view and export audited; least privilege; periodic access review	Audit coverage tests
Malicious file upload	T/D	Extension and magic-byte checks, size and row caps, no formula evaluation, CSV-injection-safe exports	Upload rejection tests
Audit-log manipulation	R	Hash chain; insert-only DB role; periodic anchoring of chain head to WORM storage	Chain-tamper test
Secrets in source control	I	Secrets from vault/env; generated locally for demo; .env.example only	Secret scanning in CI
Insecure backups	I	Encrypted backups, separate keys, restore drills	Quarterly restore test
Corruption during concurrent updates	T	Transactions, constraints, version checks	Constraint tests
Registry outage or spoofing	S/D	Mutual TLS, circuit breaker, UNVERIFIED state, re-verification	Adapter-unavailable test

24 Privacy and authorization model

Capability	AICTE admin	AICTE analyst	Inst. nodal officer	Inst. staff	Verifier	Auditor	Student
Upload / correct records	–	–	own	own	–	–	–
Search / view records	all	all (masked)	own	own	–	–	self
Set approved intake, rules	yes	–	–	–	–	–	–
Declare submission complete	–	–	own	–	–	–	–
Resolve capacity / conflict flags	yes	–	respond	respond	–	–	–
Resolve own data-quality flags	yes	–	own	own	–	–	–
Issue signed record	–	–	own	–	–	–	–
Revoke signed record	yes	–	own	–	–	–	–
Request verification	–	–	–	–	yes	–	–
Consent to verification	–	–	–	–	–	–	self
Dashboards / reports	national	national	own	own	–	–	–
Audit trail / chain check	yes	–	own activity	–	–	yes	–

- **Authentication and authorization are separate:** a valid session says who you are; every route then checks role and institution scope against the database user.
- **Consent:** employers need the student's consent; AICTE processing for regulatory purposes and statutory or investigating bodies follow the NAD exception pattern [6], subject to legal confirmation (Q1).
- **Notice:** institutions present a consent/notice text at admission describing what is shared with AICTE, why, and how to raise a grievance, aligned with DPDP Rule 3 [8].
- **Minors:** some diploma entrants are under 18; verifiable parental consent requirements [8] apply to any consent-based processing (Q2).
- **Minimisation:** no Aadhaar numbers, no addresses, no document scans; social category optional and never disclosed outside AICTE and the institution.
- **Pseudonymisation:** APAAR IDs appear in logs and audit events only as keyed HMAC pseudonyms.
- **Legal compliance is not claimed** by the existence of these controls; see open questions.

25 Dashboard and UX workflows

Workflow	Steps as the user sees them	Design choices
Institution submission	Sign in → dashboard shows intake vs admitted per programme → download template → upload CSV/XLSX → preview with ready / review / must-correct counts and seat projection → correct rows inline or leave them out → commit → flags to resolve → declare submission complete → export report	One page per task; original values kept beside corrections; consequences stated before commit ("Committing will exceed approved capacity; records are still accepted and AICTE is flagged")
AICTE monitoring	Sign in → national register with filters (year, state, discipline) → programmes over capacity shown as seat strips → open a flag → evidence → request clarification → institution responds → resolve with note → export	The seat strip makes capacity legible at a glance: filled, vacant, over-intake and each supernumerary category against its cap
Review priority	Ranked institutions with every contributing signal and its points	Disclaimer that a score is not evidence of wrongdoing
Verification	Paste token or scan QR → checklist of passed/failed checks → minimal subject details	No sign-in needed for presented tokens; tamper demo for evaluators
Student	See own record → approve or decline verification requests → copy signed record or open certificate	Disclosed fields listed before consent

Principles for a seamless workflow

- **Three steps to submit:** upload, fix what is flagged, commit. The dashboard always shows the next action and the deadline.
- **Never retype:** the template is pre-filled with the institution's programme codes; corrections happen inline in the preview; one fix can be applied to every row with the same problem.
- **Every message says what to do:** errors name the field, the reason and the fix, with a suggestion when one is certain enough to offer.
- **Consequences before commitment:** the seat projection shows what committing will do to each programme before the button is pressed.
- **Nothing lost:** previews are saved server-side, so staff can stop and resume; partial commits are safe to retry.
- **Works where institutions are:** Excel with drop-downs, browser pre-checks for slow connections, phone-width layouts for students and verifiers, Indian-language labels on the roadmap.
- **Verification without accounts:** a QR scan or pasted record is enough for an employer; consent requests reach students as notifications.
- **Accessible by default:** keyboard navigation and visible focus, WCAG 2.1 AA contrast, clear separation of errors, review flags and notes, and no chart that is not computed from records.

26 Validation and anomaly detection strategy

Deterministic rules come first; every finding carries a stable rule ID, severity, field, plain-language message, supporting evidence, suggested remediation, detection time and, once persisted, a review history. AICTE can disable a rule or change its severity, except core integrity rules, and every change is audited.

Rule IDs	Detects	Severity
R001–R002	Required fields, formats and code lists	Error
R003 / R004	APAAR missing / pending with documented reason	Error / Review
R005	APAAR malformed (12-digit rule)	Error
R006 / R007 / R008	Registry: not found / demographic mismatch / unavailable	Error / Review / Warning
R010 / R011	Impossible dates / unusual dates	Error / Warning
R012	Programme or intake not found for institution-year	Error
R013	Seat category inconsistent with nationality	Error
R014	Row year differs from upload year	Error
F001 / F002	Duplicate APAAR / roll number in file	Error
F003 / F004	Possible duplicate person in file / repeated file	Warning
D001 / D002	Updates existing record / roll number belongs to another student	Warning / Error
D003	Active enrolment at another institution	Review
D004 / D005	Attribute conflict / possible duplicate identity	Review
C001 / C002	Admissions over intake / supernumerary category over cap	Review

Review-priority score (explainable statistics)

Institution-years are ranked for attention by a weighted sum of observable signals, each listed with its evidence: seats over intake (up to 35 points), supernumerary over cap (15), unresolved cross-institution conflicts (20), share of students not registry-verified (15), share of uploaded rows still in error (10), enrolment jump of more than 30% without a matching intake change (10), and changes after declaring submission complete (10). Bands: low <25, medium 25–49, high ≥50.

Peer-group statistics

For each institution-year the platform computes features (within-intake utilisation, supernumerary share, NRI share, share of admissions dated outside the admission window, share of records changed after the completion declaration, and the largest share of students sharing one date of birth) and compares each with peers of the same institution type, discipline mix and state using a robust z-score, (x − median) / (1.4826 × MAD). Features beyond |z| = 3.5 add points with the sentence "utilisation 131% vs peer median 92%". Peer groups smaller than eight fall back to the national group of the same type.

Identity resolution

Candidate pairs are blocked on date of birth (and day/month swap) to keep comparisons small, then scored with log-likelihood agreement weights: normalised name tokens (exact, phonetic or initial-compatible), date of birth, gender, and whether the institutions are linked by a transfer. Pairs above an upper threshold become

D005 review flags; pairs between thresholds are listed for optional review; reviewers record "same person" or "different people", and those decisions become labelled data for tuning the weights. Identities are never merged automatically.

Why not machine learning yet: there are no labelled outcomes, so a model could not be evaluated, and false positives against public institutions carry real cost. A supervised model becomes reasonable after two cycles of reviewer decisions provide labels; it would need precision targets set with AICTE, per-feature explanations, and a rule that a score alone never triggers action.

27 Scalability and performance

Concern	Approach
Admission-season write bursts	Stateless app replicas autoscale; validation is batched per file; registry checks run concurrently with a bounded pool and cache within a job; uploads above a threshold move to background workers with progress polling
Large files	Streaming parse; 5,000-row cap per file in prototype, 50,000 with background processing; row-level storage enables resumable correction
Hot duplicate lookups	Unique index on APAAR ID; composite index (name_key, date_of_birth); per-job in-memory maps for file rules
Dashboards at national scale	Read replica; materialised seat-position table maintained on commit (incremental), refreshed nightly for safety; filters push down to SQL
Data growth	Partition enrolments and imports by admission year; archive closed years to cheaper storage
Public verification traffic	Verification module extractable behind its own autoscaling group and cache; signature checks are CPU-cheap (Ed25519)
Rate limits	Moved from in-process to Redis/gateway in production

28 Observability, backup and recovery

Area	Design
Logs	Structured JSON with request ID, actor role, institution ID; no names, DOB or raw APAAR IDs
Metrics	Upload volumes, validation latency, rule hit rates, registry adapter latency and error rate, verification rate, 4xx/5xx
Tracing	OpenTelemetry spans across API, engine, adapters and workers
Alerts	Registry error rate, spike in ACCESS_DENIED, audit-chain verification failure, replication lag, failed backups
Backups	Continuous WAL archiving plus nightly encrypted base backups to the DR region; 35-day retention; quarterly restore drills
Objectives	RPO 0 for zone failure (synchronous standby), 15 minutes for region failure; RTO 1 hour (assumptions to agree with AICTE)
Audit integrity	Daily chain verification job; chain head anchored to write-once storage

29 Deployment and infrastructure plan

Environment	Purpose	Notes
Local	Development and hackathon demo	SQLite or PostgreSQL via Docker Compose; synthetic seed; simulated adapter
Staging	Integration and security testing	PostgreSQL, Redis, object storage; masked synthetic data only; DAST and load tests
Pilot	10–20 volunteer institutions, PGDM first (already submit detailed data)	Live adapter if approved; otherwise manual APAAR re-check queue
Production	National rollout	Government cloud in India, two zones + DR region; WAF; KMS/HSM; SIEM

CI/CD: lint, type check, unit and API tests, dependency and secret scanning, container image signing; blue-green deploys with database migrations applied forward-only (Alembic). Infrastructure as code (Terraform) for staging and production.

30 Testing and acceptance criteria

Area	Tests	Acceptance criterion
Seat rules	0 enrolled; exactly at intake; intake+1; NRI within intake; supernumerary excluded; cap and cap+1; non-transferability; rounding	All boundary cases pass; dashboard totals equal record counts
Validation	Each rule positive and negative; registry unavailable; repeated file	Every rule fires with its ID and message
Import recovery	Partial commit, retry, concurrent commit, excluded rows, re-upload	No duplicate enrolments; remaining errors correctable
Isolation	List, detail, write, export, issue access across institutions	All foreign access returns 403/404 and is audited
Credentials	Valid, altered payload, unknown key, garbage, revoked, superseded, expired	Only untampered, active records verify
Consent	No-match indistinguishable; deny → 403; approve → fixed field list	No disclosure without consent
Audit	Events for every write and denial; tamper one row	Chain verification reports the first broken event
Integration	Bad key, missing scope, idempotent replay, changed body	401 / 403 / same response / 409
UI	Headless browser tour of every role; keyboard-only pass; mobile width	No console errors; all pages usable at 390 px
Load (staging)	5,000-row upload; 200 concurrent dashboard users; 100 verifications/s	Meets NFR performance targets

31 Prototype implementation scope (40–50%)

The prototype is built as complete vertical slices rather than a broad shell. Each slice goes from UI to API to database and is covered by automated tests.

Slice	In the prototype	Deferred
Identity and access	Login, 7 roles, institution scope on every route, lockout, rate limits, denied-access audit	SSO, MFA, password reset, user administration UI
Registry	Institutions, programmes, approved intake per year, derived supernumerary caps, submission declaration	Live intake feed from the approval portal
Ingestion	CSV and XLSX upload, Excel template with drop-downs, preview, seat projection, inline correction, bulk fix, 'did you mean' APAAR suggestions, leave-out, partial commit, retry, error report, integration API with idempotency	Browser-side pre-validation; background processing for very large files
Seat ledger	Seat-check API, live projection, server-sent events to dashboards	Admission-software plug-ins
Identity resolution	Indian-name-aware matching with explained weights, reviewer decisions	Weight tuning from decisions
Validation	Full rule catalogue (25 rules), configurable severity, simulated registry adapter	Live registry, re-verification worker
Conflicts and capacity	Cross-institution conflict and excess-intake flags with evidence, redaction, review actions, automatic clearing on condition change	Notification by email/SMS
Records	Search, filter, sort, paginate; detail with history; corrections with reason; status transitions	Bulk status updates; merge of duplicate identities
Dashboards and reports	National register, peer-group review priority, institution dashboard, four CSV reports	Scheduled report delivery; Indian-language UI
Credentials and verification	Ed25519-signed selective-disclosure records, certificate view with APAAR ID and QR, public verification, tamper detection, revocation, consent-based verifier flow, student correction requests	Bitstring status list, DigiLocker publishing, degree-body feed
Audit	Hash-chained log, chain verification, export	WORM anchoring

Demonstration scenarios

1. **Demo A, student data ingestion.** Kaveri Institute uploads a 16-row synthetic file. The preview shows 5 ready, 2 review and 9 must-correct rows (missing, malformed and unknown APAAR IDs, a future date of birth, an Indian national on an FN/OCI seat, an unknown programme, an invalid gender, a duplicated APAAR ID). The mistyped APAAR ID gets a "did you mean" suggestion that matches the student's name and date of birth. Staff accept it, fix the rest inline, leave out the duplicate and commit; the AICTE dashboard open in another window updates within seconds.
2. **Demo B, cross-institution conflict and excess intake.** Godavari's admission desk first runs a seat check, which warns that the programme is full. Godavari, already at 120/120, then uploads five students anyway, including one already active at Kaveri under a slightly different spelling. Commit opens an excess-intake flag

(+4), a TFW over-cap flag and a cross-institution conflict. AICTE requests clarification; Kaveri records the transfer; Godavari responds; AICTE resolves with a note. No record is labelled fraudulent.

3. **Demo C, secure verification.** Kaveri issues a signed record; the student shares it disclosing only institution, programme and status; a verifier confirms authenticity; editing one field breaks the signature; revocation makes it fail; a verifier without consent gets nothing; another institution's staff cannot open the record. The student then raises a correction request about a misspelt name, which appears in Kaveri's queue.

32 Limitations and external dependencies

Dependency or limitation	Effect	Mitigation
No official APAAR/ABC API access	Identity checks are simulated	Adapter boundary; labelled SIMULATED; production refuses simulated mode
Approved-intake feed from the AICTE approval portal	Intake entered manually by AICTE admin	Adapter planned; intake table already per programme-year
Supernumerary percentages differ across handbook editions [2][3]	Caps could be wrong for some programmes	Rule source stored per cap; AICTE can override; open question Q3
Legal basis and consent model under DPDP [8]	Consent rules for regulatory processing unconfirmed	Consent flow built; legal review before pilot
DigiLocker issuer onboarding	Signed records are platform-issued only	VC-shaped payload eases later publishing
Prototype uses SQLite locally	No concurrency testing at scale	SQLAlchemy keeps PostgreSQL drop-in; staging on PostgreSQL
Synthetic data only	Real-world data quality unknown	Pilot with PGDM institutions

33 Phased roadmap from prototype to production

Phase	Duration (indicative)	Outcomes	Exit criteria
0. Hackathon prototype	SIH finale	Three demo scenarios working end to end; architecture and tests	Demos run from a clean setup; tests pass
1. Hardening	8–10 weeks	PostgreSQL, Redis, object storage, MFA, background workers, re-verification, OpenAPI contract freeze, security testing	ASVS L2 checklist; load targets met in staging
2. Pilot	One admission cycle	10–20 institutions starting with PGDM; live APAAR adapter if approved; helpdesk; feedback on templates	≥95% rows accepted after correction; flags resolved within SLA
3. National rollout	Next admission cycle	All AICTE-approved institutions; intake feed; notifications; Indian-language UI	Submission completeness ≥98% by deadline

Phase	Duration (indicative)	Outcomes	Exit criteria
4. Ecosystem	Ongoing	DigiLocker publishing, degree-body feed for certificates, Bitstring status list, analytics extensions	Credentials verifiable in DigiLocker; certificates carry verified APAAR IDs

34 Risk register

Risk	Likelihood	Impact	Mitigation	Owner
Registry API unavailable or delayed	High	High	Simulated adapter for demo; UNVERIFIED state and re-verification in production	Integration lead
Institutions submit late or partially	Medium	High	Preview and correction UX, templates, API, deadline reminders, completion declaration	Product
False positives damage trust	Medium	High	Review severity, human resolution, evidence shown, rule tuning	AICTE review lead
Wrong seat rules	Medium	High	Configurable caps with rule source; AICTE sign-off on rules	Domain owner
Cross-institution data leakage	Low	Very high	Scope checks on every route, redaction, isolation tests, pen test	Security
Signing key compromise	Low	Very high	KMS/HSM, rotation, revocation of affected records	Security
Admission-season overload	Medium	Medium	Autoscaling, background processing, load tests	Ops
Foreign nationals without APAAR	High	Medium	Pending reason, review flag, later APAAR attach	Product
Module boundaries erode	Medium	Medium	Import-lint rules, module-owned tables, reviews	Tech lead
Legal basis unclear	Medium	High	Early legal review, notice text, consent records	AICTE legal
Audit log tampering by insiders	Low	High	Hash chain, insert-only role, WORM anchoring	Security
Low digital capacity at small institutions	Medium	Medium	Simple templates, Excel support, helpdesk, offline preparation	Product

35 Assumptions and open questions

#	Question	Current assumption
Q1	What is AICTE's legal basis under DPDP for collecting student-level data, and does verification by statutory bodies need consent?	Regulatory purpose for AICTE; consent for employers; statutory exception as in NAD [6]
Q2	How is consent handled for students under 18 (some diploma entrants)?	Parent/guardian consent captured by the institution at admission
Q3	Which supernumerary percentages apply per discipline and level for 2026-27, given differing editions [2][3]?	15% UG/diploma engineering, 25% PG engineering and management; configurable
Q4	Are percentage caps rounded down, up or to nearest?	Rounded down
Q5	Do lateral-entry, PwD, EWS or J&K schemes need separate capacity categories?	Modelled as SN_OTHER caps configured per approval
Q6	Is the APAAR ID guaranteed unique and permanent, and is there an official check digit?	Unique; no checksum applied
Q7	What may an institution do when a student has no APAAR ID?	Record with a documented pending reason; flag for review

#	Question	Current assumption
Q8	Should withdrawals after the admission cut-off still count against intake?	No; seat counts use current status
Q9	Who may correct a student's master identity (name, DOB) once two institutions hold records?	Owning institution with reason; conflicts go to AICTE
Q10	What retention periods apply to enrolment and audit data?	Enrolment for life of the academic ID; audit 1 year online, 7 archived
Q11	Will AICTE publish an approved-intake API?	Manual entry until then
Q12	Can platform-signed records be published to DigiLocker as an issuer?	Roadmap after issuer onboarding

36 Abbreviations

Term	Meaning
AICTE	All India Council for Technical Education
APAAR	Automated Permanent Academic Account Registry (12-digit academic ID)
ABC	Academic Bank of Credits
NAD	National Academic Depository
NRI	Non-Resident Indian
OCI	Overseas Citizen of India
FN	Foreign National
CIWG	Children of Indian Workers in Gulf countries
TFW	Tuition Fee Waiver scheme
PGDM	Post Graduate Diploma in Management
VC	Verifiable Credential (W3C)
JWS	JSON Web Signature
KMS / HSM	Key Management Service / Hardware Security Module
DPDP	Digital Personal Data Protection Act 2023 and Rules 2025
RBAC	Role-based access control
RPO / RTO	Recovery point / recovery time objective
WORM	Write once, read many storage

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